

MANUAL

A field guide to commercially produce low-cost, high-quality novel protein source to supplement feeds for poultry, pig and fish industries and the valorization of organic by-products



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Introduction

The black soldier fly *Hermetia illucens* (Linnaeus, 1758) (Diptera: Stratiomyidae) is a very adaptable saprophagous species. The larvae of *H. illucens*, the Black Soldier Fly (BSFL), possess many qualities that make them attractive for converting urban food waste into animal feed and other valuable products. They have been successfully fed to a variety of livestock animals including pigs, tilapia, and catfish. BSFL have been shown to effectively digest a wide range of organic waste products such as offal, kitchen waste, and fruit and vegetable waste (Nguyen et al. 2015). BSFL are also useful for managing manures and when grown on chicken manure were able to reduce the amount of manure by 50% and the larvae had a resulting composition of 42% protein and 35% fat (Sheppard et al. 1994). The conversion of low value manure, or other organic waste streams, to bulk proteins and lipids creates many opportunities for developing valuable products. The ability of the BSFL to create such high proportions of lipids has even attracted attention to how their culture might be applied to biofuel systems (Li et al. 2011). Despite their fondness of manure, BSFL are thought of as a non-pest species (Sheppard et al. 1994) and have been shown to reduce *E. coli* levels in chicken and dairy manures (Erickson et al. 2004; Liu et al. 2008) as well as compete with house fly larvae, thereby decreasing their population (Bradley and Sheppard 1984).

The ease of harvesting the BSFL is also of immense value. The last stage of the BSFL's life cycle before pupation is a migratory stage where the larvae develops a large fat store, empties its gut, and then seeks a place away from the waste to pupate (Sheppard et al. 1994; Newton et al. 1977). This makes collecting the larvae as simple as providing a singular high point that falls into a collection bucket. Being able to harvest the larvae passively makes scaling systems much simpler and much more cost effective than more intensive systems. After the food waste has been processed by the BSFL, the leftover biomass is nutrient rich, low-odor, humus that can be added to soils as an amendment to increase organic matter and fertility (Diener et al. 2009).

One drawback of BSFL when compared to fishmeal is the presence of chitin that forms the BSFL outer membrane. Chitin is largely indigestible by fish and it should also be noted that the nitrogen content of chitin is 6.89%, which should be taken into account when formulating feed as it significantly changes the amount of digestible crude protein (Diener et al. 2009). Studies have shown that BSFL are composed

of 42.1 - 50% protein depending on their rearing substrate. Encouraging the use of BSF farming (**Figure 1**) among the African farmers including the youth and the women can be a model of business incubation that will support various fields of development in an organic and chemical free environment. Therefore, the aim of this manual is to guide farmers to produce their own insect proteins using BSF larvae and to use the residue as an organic fertilizer for their crops.



Figure 1: Adult black soldier fly

Economic Importance

In the last few decades there has been considerable interest in using larvae of *H. illucens* for organic waste control, composting (Sheppard et al. 1994; Lalander et al. 2015), and as a high-quality food supplement for poultry (Tomberlin et al. 2002), fish (Bondari and Sheppard 1981) and swine (Newton et al. 1977) globally. *Hermetia illucens* has been found to be highly resistant to insecticides (Turchetto 2000; Turchetto et al. 2001) and tolerate a wide range of pH (Brues 1928), thus making them winning competitors and predators of autochthonous flies as observed on remains in fields treated with pesticides. Black soldier fly larvae have been widely used to recycle animal tissue (Tomberlin et al. 2005) and other waste products (Sheppard et al. 1994). Because of this ability, the black soldier fly is a recognized method for sustainable recycling of animal wastes (Sheppard et al. 1994), feces (Myers et al. 2008), and plant materials (Tomberlin et al. 2009).

Larvae of the black soldier fly can reduce dry matter in such waste by 50% (Sheppard et al. 1994) and house fly, *Musca domestica* L. (Diptera: Muscidae) populations by 94 - 100% (Bradley and Sheppard 1984), although producing viable feedstuff for livestock, poultry (Sheppard et al. 1994), and aquaculture (St. Hilaire et al. 2007). Unlike house flies, soldier fly larvae can be used for the production of biodiesel (Li et al. 2011). They have also been widely observed to inhibit oviposition in house flies *Musca domestica* L. (Diptera: Muscidae) (Furman et al. 1959, Sheppard 1983; Bradley and Sheppard 1984), to recycle organic matter (Larde 1990) and reduce the bacteria *Escherichia coli* and *Salmonella enterica* in chicken manure (Erickson et al. 2004; Liu et al. 2008) and total nitrogen concentration by 62%, thus methane formation and off-gassing. Larval and bacterial activities can reduce the dry matter, but also other components such as nitrogen or phosphorus (Oliveira et al. 2016). Experiments with cow manure showed reductions of 43% in nitrogen and of 67% in phosphorus (Gobbi 2012).

This explains why, they have been widely used as biological control agent for house flies in commercial poultry operations and stables (Saccà 1964; Skuhravá et al. 2010). Black soldier fly thrives in large numbers on organic waste and therefore can be considered as a solution to environmental problems associated with organic waste in our cities, markets, breweries but also our farms. As the larvae mature they crawl out of the basin, thereby self-harvesting themselves, and are subsequently available as livestock feed. In addition to being a good source of oil and protein for animal feed, black soldier fly larvae have the potential of improving organic waste into a rich fertilizer.

Nomenclature

The Black Soldier Fly is an insect that belong to the Stratiomyidae family (Diptera), which comprises 260 known species in North America (Triplehorn 2005). It was originally described as *Musca illucens* Linnaeus, 1758. The species was successfully changed to *Musca leucopa* Linnaeus, 1767. The genus was later changed to *Hermetia rufiventris* Fabricius, 1805. The genus was later placed in a new species *Hermetia pellucens* Macquart, 1834; *Hermetia nigrifacies* Bigot, 1879; *Hermetia mucens* Riley & Howard, 1889. The correct name of the species is now *Hermetia illucens* var. *nigritibia* Enderlein, 1914 and then *Hermetia illucens* Copello, 1926. Of the five known genera of this subfamily three are restricted to the Neotropical Region and one is Australasian. The remaining genus *Hermetia* Latreille, 1804, is by far the

most speciose with 76 species, and more widely distributed, having representatives in the Nearctic (12), Neotropical (52), Afrotropical (3), Australasian (10) and Oriental Regions (14) (Woodley 2001).

Distribution

The black soldier fly is believed to be native to many areas including most of the Western Hemisphere and the Australian region from Samoa to Hawaii, throughout South America (Medellín, Colombia) and North America as well as Asia (Carles-Tolra and Andersen, 2002). Although it has been suggested that *H. illucens* might have been first brought to Europe around 500 years ago (Benelli et al. 2014), the first verifiable Palaearctic record of the species is from southern Europe (Malta) in 1926 (Lindner 1936, Venturi 1956). The subsequent spread and most recent records have been mainly along the Mediterranean coast of Spain, France, Italy and Portugal in the 1950s and 1960s (Leclercq 1969, 1997; Carles-Tolrá et al. 2002). It is believed to have invaded France in 2007, when the first record was reported in Rillieux-la-Pape (Rhône), thereafter, several other population of the flies were reported in Saint-Maurice-de-Beynost (Ain), Villeurbanne (Rhône), Sainte-Foy-lès-Lyon (Rhône) and Bourg-lès-Valence (Dauphin 2003; Alain 2006; Richoux 2009). Although, reported in Italy in 1956 (Venturi 1956), some authors suggest that only in recent years has it become widespread throughout Italy with populations observed further in the north, up to 1300 m above sea level (Mason et al. 2009).

Additional, reports have demonstrated their occurrence in Switzerland, Croatia (Rozkošny and Knutson 2007) and Slovenia [Ljubljana; Vic - Ljubljana; Prade near Koper and Božmarin] (Groot 2011). Groot (2011) strongly believed that the species entered Slovenia by dispersal via Italy after the initial introduction in Europe (Turchetto and Vanin 2010). In more recent years, the species has been documented spreading northwards in Central Europe. Ssymank and Doczkal (2010) recorded it from Germany, while Roháček and Hora (2013) recorded it from the Czech Republic. The first records from Malaysia, Hawaii, Solomon Islands, New Caledonia, Mariana Islands, Palau, and Guam was reported in the 1940s. The apparent spread of this species along coastlines and islands suggests that maritime transport may have played a role in repeated accidental introductions. This is further supported by the observation put forward by Venturi (1956), Saccà (1964) and Skuhrová et al. (2010), who presumably reported that the primary pathway of introduction of *H. illucens* throughout the world was through deliberate importation during the Second World War by U.S. troops as biological control agents aided by globalization of trade and poor quarantine infrastructure during this period. Despite the widespread distribution of *H. illucens* in Africa, the only published record comes from South Africa as early as 1915. The occurrence of *H. illucens* in several parts of the world around the same time, clearly demonstrates that the exact original distribution of *H. illucens* might not be well known, although it cannot be excluded that it originally occurred in the Southeastern United States as far back as 1800s and its current North American range seems to reflect a northward spread from a native range in Central America and the northern parts of South America in historical times. Although, after World War II, *H. illucens* became widespread in the western part of the Mediterranean Sub-region, interestingly, there seems to be no record from the Mediterranean areas of North Africa. Several questions remain unanswered about the history and current status of the *H. illucens* in the world today because the available data does not inform us of how and when it arrived as well as the relationship between the different populations across the world (Figure 2).

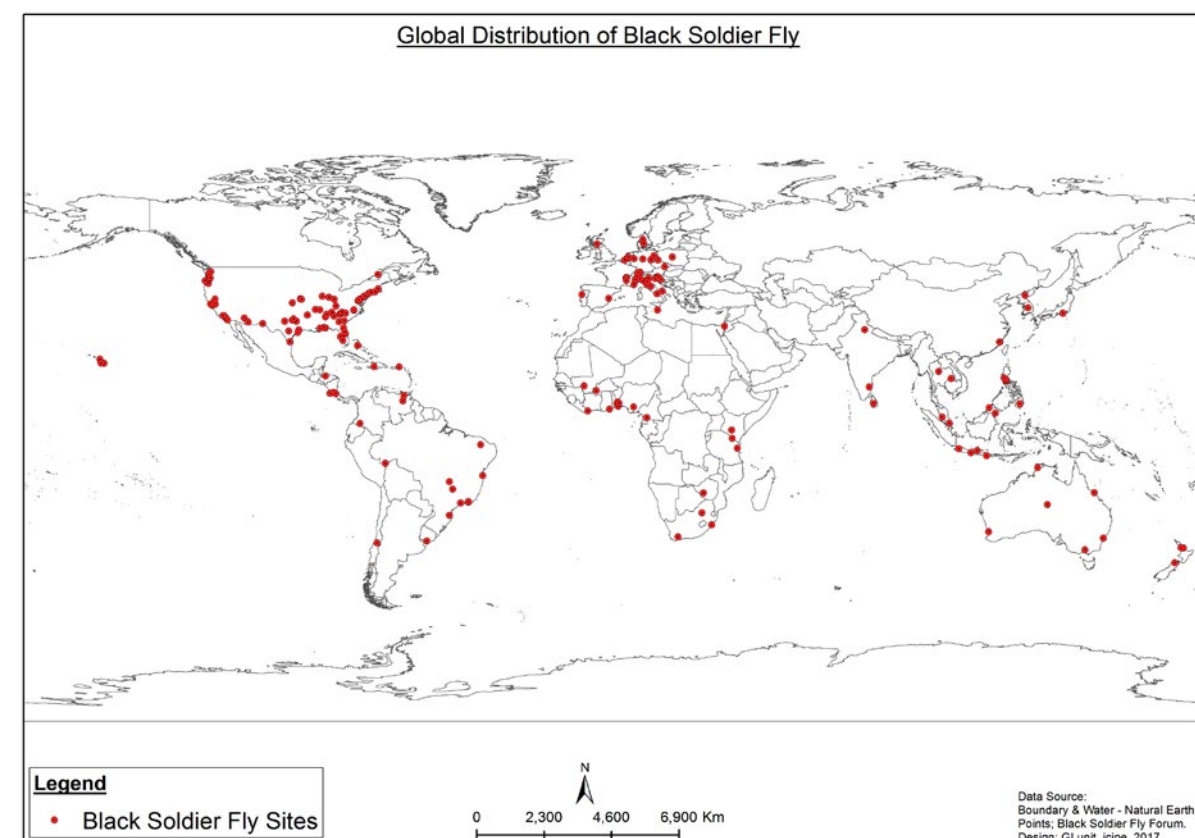


Figure 2: Global distribution of black soldier fly

Biology of black soldier flies

Adults

The head of adult flies is small and narrower than the body. The eyes are broadly separated in both sexes. Members of the soldier fly family Stratiomyidae can range in color from yellow, green, black or blue, with some having a metallic appearance. Many are mimics of other flying insects, such as bees and wasps. The black soldier fly adults are sleek looking fly with a wasp-like appearance and are black or blue in color (Figure 3). Soldier flies also have two translucent “windows” located on the first abdominal segment. Adults range from 15 to 20 mm in length. (Sheppard et al. 2002). The adult’s antennae are elongated with three segments, and legs have white coloration near the end of each leg. However, like most flies, the black soldier flies only have two wings (wasps have four) and does not possess a stinger. Although the loud buzzing they create when flying is enough to concern many people, adult soldier flies pose no danger. A beneficial fly because unlike most other flies, BSF adults do not go into houses. Black soldier fly do not have functional mouth parts to chew their food, they have a sponge-like mouth part that allows them to lap up liquid. If the food is already in liquid form, then they simply places the sponge on the liquid but if the food is in solid form, the insect has to first regurgitate salivary secretions onto the solid food, allowing it to liquefy before it feeds similarly to other *Musca*. Therefore, they are not associated in any way with the transmission of disease. Adults do not bite, bother or annoy humans. Their activity is limited to mating and egg-laying. Note that only females visit waste. Males never come near waste, since males do not lay eggs. When females visit waste to lay eggs, they do not come into contact with the waste. They lay their eggs above or to the side of the waste, never on the waste, otherwise they run the risk of their eggs being eaten along with waste.



Figure 3: Adult male (left) and female (right) black soldier flies.

Mating and Oviposition

Two days after adult emergence from the pupal case, mating can occur. A male black soldier fly intercepts a passing female in mid-air and they descend in copula (Tomberlin and Sheppard 2001). Male soldier flies utilize lekking sites, where they await female soldier flies. The lekking sites attracts virgin females seeking mates and these sites are usually defended against other male soldier flies. When a male intrudes upon the territory of a resting male, the resting male seizes the intruder. After a brief descent, the invading male will retreat.

BSF mating begins in the air with aerial questing after stimulation by light (Furman et. al 1959). Aerial questing is thought to be an important aspect of the mating process, and any facility should have enough air volume to facilitate this event; however, no recommendation on volume and adult fly densities has been reported in the literature. Mating is generally on or closely to the ground, with the males and female facing different directions (**Figure 4**) or male mounted on the female (**Figure 5**). However, it is always very difficult to achieve successful mating or egg collection in cages measuring less than 1 m² under artificial colony, thus larger cages are needed to increase the probability of successful mating. Cages holding adults of 2 x 2 x 4 m, 1.5 x 1.5 x 3 m or 3 m³ that allows free flying inside ensures significant mating occurring. It is also difficult to get flies to mate with only a heated 66 m³ greenhouse. Several observations have shown that plants should be included inside the cages as lekking sites, though not always essential for mating to occur, so long as other conditions are suitable.

Another requirement for successful mating is strong light, preferably sunlight. Sheppard (2002) reported that a minimum of 63 $\mu\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$ was required for mating with optimal results at more than 200 $\mu\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$. Although, Zhang (2010) reported that most mating occurred in the early morning, Kim et al. (2008) found that mating was most frequent during the day, when light was most intense (**Figure 6**). Light also affects oviposition, with most occurring when the light is brightest. Studying the behavior in wild populations, Booth and Sheppard (1984) found a clear peak in oviposition between 13:00 – 17:00 pm. Similar results were reported by Kim et al. (2008) for laboratory reared laboratory based colony (Fig.5).



Figure 4: Males and female black soldier fly facing different directions during mating



Fig 5: Males black soldier fly mounted on the female during mating

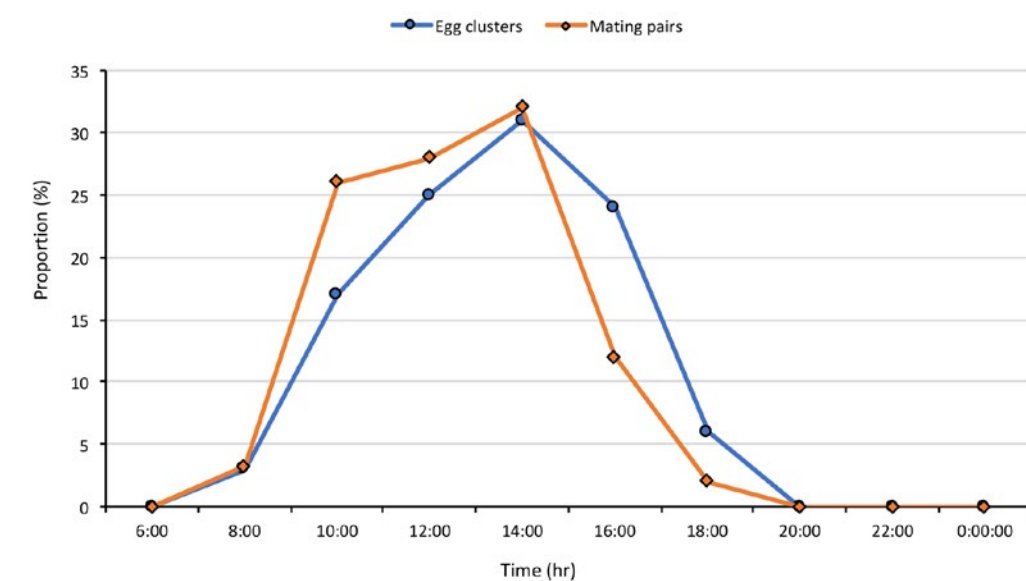


Figure 6: Active mating period of black soldier flies in rearing facilities and total number of egg clusters oviposited per day.

Egg laying usually occurs two days after mating. Females seek out an area that is close to a food source to deposit their fertilized eggs. The mechanism for this action is believed to be the detection of volatile chemicals from rotting wastes (Sheppard et. al. 2002). Females will also leave chemical markers that attract other females to a suitable egg laying site. Females prefer not to lay eggs directly on a food source but near it. The ideal egg laying site should be maintained at 27°C with an ambient relative humidity of 60% or more: at these conditions egg hatching rates of 80% or more have been observed (Holmes 2010, Sheppard et. al. 2002).

A female produces about 900 eggs in her short life of 5 when no water is provided to above 40 days when provide with 50% sugar solution. In most BSF rearing facilities, the egg laying and waste processing activities are conducted in separate locations. Eggs are usually collected in the corrugations of cardboard, or flutes, and then transferred to the hatching trays with suitable substrates. This is done because eggs are fragile, small and vulnerable to changes to environmental variables; the flutes provided protection for eggs and encouraged the female's ovipositor to lay eggs in the confined space. In a waste management facility, it would be preferable to automate or remove the need for egg handling.

Eggs and method of eggs harvesting from the wild and within the colony

The natural instinct of black soldier fly is to oviposit in dry cracks close to moist, decomposing organic materials. It has been found that the small openings (flutes) in strips or rolls of corrugated cardboard make highly suitable oviposition sites for the black soldier fly, with smaller size corrugated (3 flutes per centimeter) preferred (Sheppard 2002). The dry cardboard is usually positioned vertically over an oviposition attractant, or the rubber tubes with grooves directly on the substrate or cardboard directly on the substrate with additional materials to present the cardboard from getting soaked in the substrate (**Figure 7a, b & c**), such as wet chicken waste or well moistened poultry layer mash for wild trapping as well as in laboratory colony.

The BSF egg laying process occurs when the females are 8 to 9 days old. Females BSF don't land directly on the waste, a single female deposits a mass of about ~500 to 1200 eggs in cracks and crevices or surfaces above or adjacent to decaying matter (rotten organic wastes) such as manure or compost, carrion, garbage, and other organic waste. Adult flies can be enticed to lay eggs in small holes over the grub bin or lay clusters of eggs in the edges of corrugated cardboard placed next to the compost. Multiple females have been observed to often oviposit within the same flute. Adult flies in the laboratory colonies have been shown to mate and lay eggs at temperatures ranging from 24 - 35°C, while in the wild 99.6% of oviposition has been reported to occur between 28 - 37°C.

The eggs hatch into larvae in about 4 days. Each oval shaped egg is about 1 mm in length, and pale yellow or creamy white in color (NCIPMI 1998). Although they can be stored at room temperature for several weeks, their longest shelf life is achieved at 10–16 °C.



Figure 7a: Adult female fly enticed to lay cluster of eggs in corrugated cardboard placed over the grub bin.



Figure 7b: Placement of rubber tubes with grooves directly on the substrate can also serve as an important oviposition sites for female flies



Figure 7c: Placement of rubber tubes with grooves directly on the substrate in laboratory colonies can also serve as an important oviposition sites for female flies

Larvae

After hatching, the larvae start consuming in a 360-degree cellular fashion. Newly hatched larvae are approximately, 1.8 mm long and are a dull white to cream colour. Optimal moisture content for the feed ranges from 60% to 90% (Myers et. al. 2008). Optimal temperatures for efficient food processing range from 27 to 33°C (Sheppard et. al. 2002). Lower temperatures are most likely tolerable because the maggots generate heat as they consume food through their writhing motions. The maggots secrete enzymes that make the food digestible prior to ingestion by liquefying the waste as they consume it. The moisture content of the resource is important as it affects BSF development. Moisture contents outside the optimum range will cause adverse effects. Too much moisture will force the maggots leave the food/resource matrix they are feeding in; and not enough will prevent efficient consumption (Fatchurochim et al. 1989). However, these moisture circumstances can be exploited to engineer a BSF waste management system: the moisture preferences of the larvae could provide a simple method to control and direct the maggot's location in the system.

They are easily distinguished from other fly larvae by their obviously segmented bodies. The distinctive row of short, bristly hairs on each of their 11 segments also help identify BSF from other larvae. The distinctive row of short, bristly hairs, on each of their segments also helps identify BSF from other larvae. They are a dull, whitish color with a small, projecting head containing chewing mouthparts. Their entire bodies are designed to consume any biomass around them. Larvae pass through five larval instars (**Figure 8**) and require approximately 13 - 18 days to complete development depending on the temperature, food type and food availability (Hall and Gerhardt 2002). During larval development, black soldier fly larvae are insatiable voracious feeders of organic waste – No limits, eats everything that begins to rot. BSF larvae are photophobic as they tend to bury themselves in the substrate when exposed to light.



Figure 8: Different larval stages of black soldier flies

This implies rearing them under shade. As such adults do not need to feed and rely on the fats stored from the larval stage (Newton et al. 2005). It may take up to six months for larvae to reach maturity due to black soldier flies' ability to extend their life cycle in hostile circumstances. The larvae can be up to roughly 27 mm in length and 6 mm in width. They have a pale white color with a small black head containing their mouthparts (Newton, 2005) (**Figure 9**).



Figure 9: 5th instar larvae harvested from the insectarium

It also appears that soldier fly larvae secrete chemicals that warn other fly species that a food source colonized by soldier fly larvae is not an ideal egg laying site leading to effective reductions of the common housefly (Bradley and Sheppard 1983). The nutrition source used in the larval stage can also affect adult fly characteristics (Tomberlin et. al. 2002). Thus, in a full-scale waste processing facility the nutritional content of the incoming waste stream may need to be monitored to ensure that the larvae are eating a balanced diet. This would be important because the facility would require healthy adults to maintain egg production at required levels.

Migration and Pupation

The wandering pre-pupal stage lasts 7–10 days at 27°C, during which time the larvae stop eating, once they have consumed enough food they begin to migrate away from the food source (Sheppard et. al. 1994). Following the maturity of the BSFL, the natural instinct is to climb up and out from the damp, decaying organic matter in which they have been living and find somewhere dry. In laboratory culture this can be exploited – escape ramps allow pre-pupae to leave the substrate mass and drop conveniently into plastic containers for collection (Figure 10). Pre-pupal stages begin where the fifth instar (final molting phase in *H. illucens*) larvae remove themselves from their feeding sites in search of a dry, sheltered environment (self-dispersal phase). Once this occurs, pupation commences. This is a significant advantage of the species, as separating larvae from their food source is one of the major barriers to using other insect larvae in animal feeds.



Figure 10: Self-harvested black soldier flies' pre-pupae leaving the substrate mass and dropping conveniently into plastic containers for collection (self-dispersal phase) at the icipe rearing facility.

Under natural conditions pre-pupae will change its colour, a circumstance that could be exploited to further sort them for quality purposes. However, the larvae are accomplished burrowers and can enter small spaces and crevices with ease and when they are wet they can attach themselves to a wide variety of surfaces including plastics, wood, rubber and metals. This can pose problems trying to contain them in specific feeding areas. It has been found beneficial to provide pre – pupae with moist wood shavings (60-70%) or other wet light material rather than forcing them to pupate on a bare or dry surface. This reduces the time spent roaming during the non-feeding pre-pupal stage, ensuring they are maximum weight during pupation and resulting in a higher rate of adult emergence.

The time spent by each pre-pupa in the migration stage varies but appears to be dependent on the maggot's ability to locate an ideal pupation site. A study suggested that the maggots favour drier conditions for a pupation site but require ambient humidity levels of approximately 60% to emerge as adults (Holmes 2010, Sheppard et. al. 2002). Migrating maggots are suspected of leaving chemical trails that other maggots follow creating a migration path.

Another characteristic of an ideal pupation site is protection from predators and unfavourable environmental conditions, such as flooding. The pupation media itself should be porous and loose to allow for easy burrowing of wandering maggots. A medium with these properties should also provide adequate oxygen levels so the pupae can breathe. If the pupation medium is too fine the spiracles, or breathing structures, can become clogged possibly resulting in death.

The depth of the pupation medium is also important. If it is too deep, the emerging flies will fail to reach the surface. If it is too shallow, the maggot may not deem the location adequate and continue to wander, wasting its fat reserves thereby reducing its harvest value or its chance to successfully mate. Wandering maggots have been shown to pupate without a pupation medium if no suitable medium is present (Holmes 2010). Studies have shown the ideal depth for a pupation medium is 15 to 20 centimeters. Pupation can last five to seven days depending on the temperature and ambient humidity.

Pupae

Before pupation, the sixth instar larvae (pre-pupae) disperse from the feeding site (substrate) to dry sheltered areas, such as ground vegetation, to initiate pupation. The exoskeleton (skin) darkens and a pupa develops within (**Figure 11**). Pupation requires about 7 – 12 days (Hall and Gerhardt 2002). It is advisable to provide the pupae with moist wood shavings (60-70%) or other wet light material rather than forcing them to pupate on a bare or dry surface to avoid desiccation, which will lead to unacceptable high level of mortality (i.e. low emergence rate).



Figure 11. Pupa of the black soldier fly, *Hermetia illucens* (Linnaeus) with darken exoskeleton (skin)

Life Cycle of the Black Soldier Fly

The BSF has five stages in its lifecycle: egg, larvae, prepupal, pupae and adult. These are shown in **figure 12**. The larval stage is further divided into phases called instars. An instar is defined as the period between each moulting of their exoskeleton. The number of instar stages varies for different fly species: *Hermetia illucens* has five instar stages. All of the adult fly's nutritional requirements are obtained during its larval stage and adult flies survive on their fat reserves obtained as maggots. When this fat reserve is depleted the adult dies (Myers et. al. 2008). The larvae have a wide range of diets: they can consume animal feces, rotten and fresh flesh, fruits, restaurant waste, kitchen waste, cellulose and possibly a variety of other organic wastes (Nguyen 2010, Holmes 2010, Sheppard et. al. 2002, Tomberlin et. al 2002). Soldier flies, from egg to adult, have an estimated life cycle of 38 - 40 days but this length depends on the environmental conditions present and the rearing diet. Waste consumption rates appear to depend on the size of the maggot and the type of food being consumed (Diener et. al. 2009).

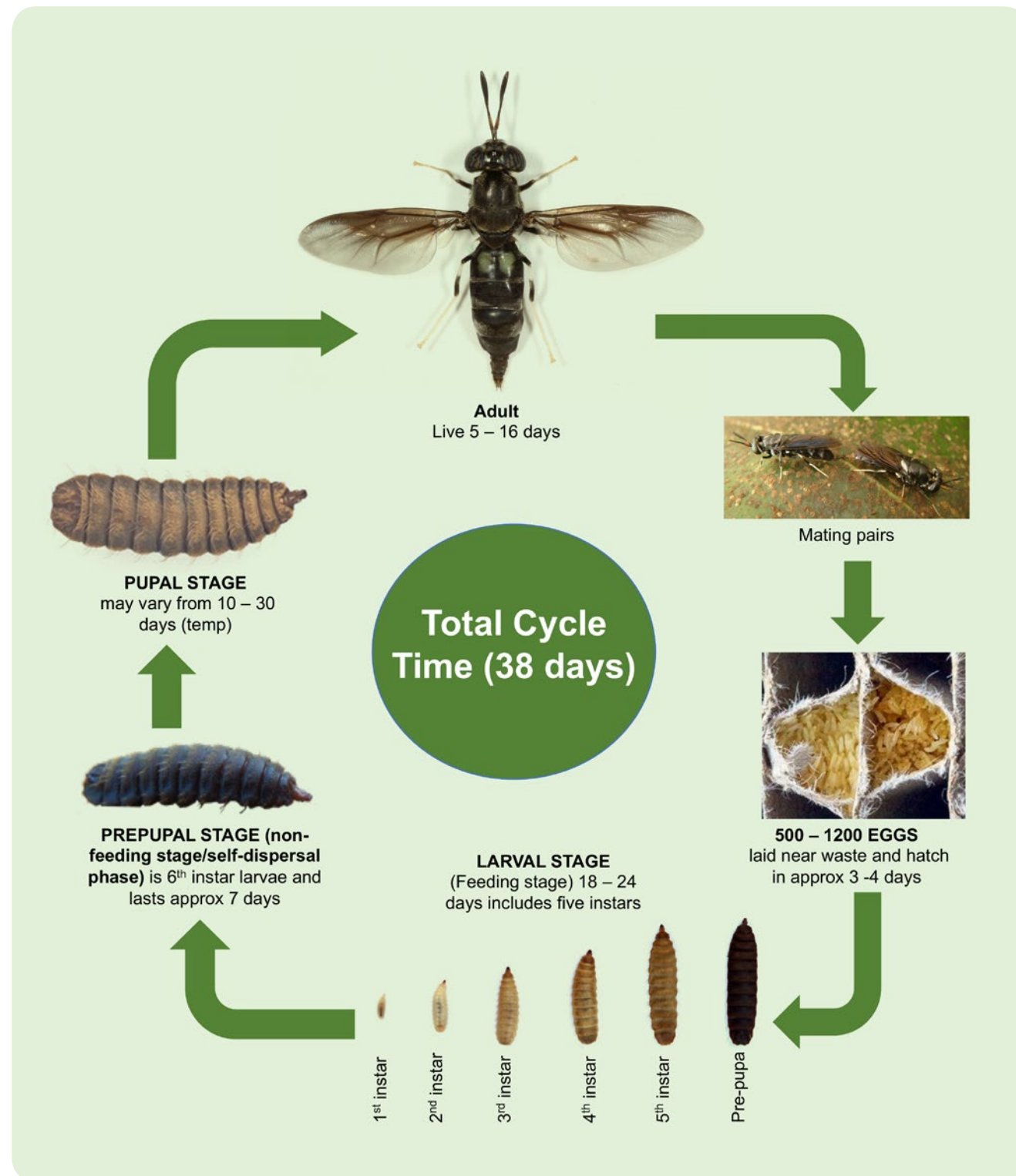


Figure 12: Life cycle of the black soldier fly

Harvesting and farming BSF

How to design an insectarium

The design of a standard BSF insectarium is presented in **figure 13, 14, 15 and 16** provides an overview of available and readily used structures in Kenya. Basically, the insectarium is a normal cage that has a 1m² basis and 1,8m high. The top roof is designed in conical form but this can also be a square design. Netting material is used to cover the wooden/plastic/metallic structure. The cover can be a mix of plastic and netting material to protect the insectarium from rains. As space is essential for the flies to mate, it is important to design the insectarium according to the ambition of the farmers (small or large production unit). In any case, the insectarium should be designed in such a way the farmer can access the inside through a clip closure or a simple opening to place new pupae, collect eggs or clean out dead adults which can be placed in the larvae bind. Below are example of different adult cage structures both for indoor and outdoor rearing purposes



Figure 13: Indoor design of a standard BSF insectarium



Figure 14: Indoor design of a dismantle rearing BSF cages suitable for smallholder farmers



Figure 15: Outdoor design of a standard BSF insectarium with a roof



Figure 16: Larger insectarium with plastic blinds to avoid rain and wind

The Larvarium

Eggs collected from insectariums are transferred to the larvarium. Eggs are about 1mm long and a creamy yellow colour in appearance. Eggs are then placed on rearing substrate to hatch (**Figure 17**) or could be stored and kept in cool places. Eggs take 4 days to hatch at optimum conditions of around 25°-30°C. Once they hatch they migrate down into the fresh rearing substrate (chicken manure, cow dung, rabbit dung, pig manure, market waste, kitchen waste etc), which are recommended as a starter diet. Avoid immersing them on highly soggy waste. The huge number of newly hatched larvae are then distributed into several other trays to avoid overcrowding in the rearing trays, which are later distributed onto wooden stands in the insectarium for onward development to pre-pupal and pupal stages.



Figure 17: Newly hatched larvae from the cardboard materials into the starter diet

Design and dimension of black soldier rearing trays and stands at the icipe rearing facility

The black soldier fly rearing room facility is equipped with wooden stands and rearing trays (**Figure 18**). The wood used to make the rearing stands are 7.5 cm x 5 cm thick (3-inch x 2 inch). The height of the bigger stand is 180 cm by 420 cm 420 cm, while that of the smaller stand is 180 cm height by 129.5 cm length. Each stand has four shelves separated from each other by 30 cm space. The width of each shelf is 66 cm with a length of 420 cm. Each shelf is equipped with two metallic troughs on either side to collect BSF larvae during the dispersal phase.

Each trough has a length of 420 cm with a depth (height) of 5 cm and 6.5 cm bottom width. Each trough is also provided with a bent of 5.5 cm by 1 cm to allow it hook with ease on the wood. At the end of each trough a 2-Liter plastic container (21 cm length x 15 cm width x 7.5 cm height) is attached on either side for collection of crawling pre-pupae from the rearing substrates.

The top surface of the metal tray used from rearing black soldier fly larvae measures 76 cm in length by 27.5 cm width with a depth (height) of 10 cm. The bottom of the tray measures 52 cm in length by 27.5 cm width, which allows for both edges of the tray to be inclined at an angle 35 °C. Both ends of each tray are provided with a collar of 5 cm long to allow it sit smoothly on the edge of each trough.

Each BSF rearing room facility holds three wooden stands. Each of the two larger wooden stands hold 8 trays per shelf making a total of 64 trays, whereas the smaller stand holds a total of 16 trays. Each of the BSF rearing room has a total of 80 rearing trays, as such the two rooms have 160 rearing trays. Each of these trays has been estimated to hold approximately 15000 - 20000 BSF larvae.

Following the maturity of the BSFL, they climb up and out from the rearing substrate and seek for drier surfaces for pupation. In laboratory culture we have exploited this behavior by providing escape ramps for the pre-pupae to leave the substrate mass and drop conveniently into plastic containers for collection.



Figure 18: Photo of the wooden stands and rearing trays in the BSF rearing room facility at icipe.

Rearing substrate

Black soldier larvae are voracious feeders and will consume an incredible range of organic materials. Regarding the type of substrate, black soldier flies can thrive on a huge diversity of organic waste streams like kitchen and market waste, brewery waste, livestock droppings or manure (pig, goat, cow or chicken manure), municipal waste (human) and even water hyacinth biomass. A substrate combining kitchen scraps and coffee grounds may help to boost their metabolism.

Before the advent of closed sheds, the space underneath poultry layer houses would become naturally heavily infested with BSF larvae, which reduced the volume of the waste by at least half (50%).

In addition to reducing the need to dispose of poultry manure, the insect can provide the additional benefit of outcompeting housefly larvae, which are otherwise major pests. BSF larvae can successfully develop in poultry and swine manure, sewerage sludge and liquid leachate from compost. They can even live in dead animal bodies; black soldier fly larvae have been used to estimate postmortem intervals for human corpses.

Not all organic materials provide a suitable diet for BSF. Newby noted that BSF populations in compost bins tended to disappear if large amounts of putrescible wastes (such as meat scraps), grass clippings or sawdust were added. Although BSF larvae fed entirely on liver or fish renderings grow rapidly, mortality is extremely high. Combined mortality from pupal, pre-pupal and adult stages was 99.8% and 98.2% for larvae fed fish or liver respectively. Meat meal is similarly considered unsuitable for rearing BSF.

While plant materials are a natural food for BSF, the larvae cannot easily survive on organic materials such as plant stems, grasses and mature leaves. It seems likely that the larvae are unable to effectively digest

lignin, which is a key component of plant cell walls and highly resistant to biodegradation. This may explain why cow manure is far less suitable for BSF than poultry or pig manure; whereas pig manure contains only about 2% lignin, cow manure can be 10% lignin and wheat straw is about 23% lignin.

After three to four weeks, pre-pupae will start crawling out of the food substrate. Pre-pupae are self-harvesting, there is a habit to leave the substrate and search out a dark, dry place to pupate (**figure 19**). This habit is often a good bonus in the production of BSF. Collected pupae are transferred for pupation to get the adult flies in the insectarium for colony maintenance.



Figure 19: Pre-pupae harvest from the insectarium for adult stock colony maintenance at *icipe* rearing facility, Duduville campus

Advantages of farming black soldier fly

- Fast growth and development
- Efficient disposal of waste materials
- Larvae self-harvesting at maturity and relatively easily processed
- Larvae are suitable for some high value uses, especially with refinement of culture and processing methods
- Frass suitable for lower value uses – saleable item
- Adult flies are short-lived and lack functional mouthparts, so no pest or disease issues
- Some commercial production now starting in many areas in the country – expertise in setting up a facility is available

Environmental Conditions Temperature- Lighting- Humidity

- Black soldier fly larvae develop easily at 70% humidity but the optimal humidity for black soldier fly mating is between 30 and 90%.
- Larvae avoids contact with light and remain 10 inches below substrates. The tray or rearing container should be around 30 cm-deep with a slanting side or ramp of 40~45-degree incline leading gutter with a bucket to collect the self-harvesting pre-pupae.
- Units placed in direct sunlight will heat up too much, resulting in early crawl-off of premature adults (lighter in color) and may even cause the entire colony to collapse.
- Even in full-shade conditions, temperatures over 35°C or too much moisture will cause premature crawl-off. The best solution is to use an inexpensive shade canopy which protects the entire colony from the heat of the sun. During extended heat waves, turning over of the food substrate is important to keep the temperature low.
- Trays can be constructed in such a way that pre-pupae will be directed to collection vessels, from which they can be removed periodically. They may then be kept at 4°C for several weeks until such time as they are required.
- The advantages of this are that the trays can be larger and less monitoring is required. Feed stock may be added and pre-pupae collected continuously until such time as the tray is full. At this time newly, hatched larvae may be added for a further few weeks. Typically, it will take around 4 to 6 weeks for larvae to fully grow and reach the pre-pupa stage, therefore the addition of new larvae should be halted around 2 months prior to cleaning out the residue feed and frass.
- Therefore, it is desirable to have more than one tray, so that continuous production can be maintained. Harvesting the BSF larvae, prior to pupation, adds complexity to the process because they will not typically self-harvest. This may be desirable, however, if the goal is to produce optimum nutritional quality of the larvae. The pre-pupae and pupae have much tougher skins with high chitin content, which is anti-nutritional.
- Therefore, to provide a high-quality feed material for livestock, the larvae must be collected before pupation occurs. The downside of this is that they must be collected in some way, either by hand or

an automated system must be used.

- Bulking out with high carbon materials to compost efficiently. Ideal composting conditions require a C: N ratio of around 30:1 and sufficient oxygen exchange. Higher nitrogen tends to lead to venting of ammonia and other nitrogen gases whereas poor oxygen availability leads to anaerobic processes and the release of methane.

Potential Uses for BSF Larvae

In addition to being voracious consumers of organic wastes, BSF larvae contain useful organic compounds that have commercial and industrial value: these include:

- 42.1 - 50% Crude protein
- 34.8% Lipids
- 7.0% Crude fibre
- 7.9% Moisture
- 1.4% Nitrogen free extract (NFE)
- 14.6% Ash
- 5.0% Calcium
- 1.5% Phosphorus

The larval excretions and pupae casings can be used as a replacement for peat moss but unlike peat moss, the casings and excretions could easily be renewable resources. The larvae have been used as a substitute for dairy, swine and poultry feeds. In this role the BSF larvae both stabilize problematic wastes and create a value-added agricultural product. The larvae have also been fed to aquaculture systems – an industry facing a potential shortfall of feed protein.

Processing

Black soldier fly larvae and pre-pupae are suitable for feeding a range of animals. It has been demonstrated that BSFL is a preferred or substitute of fishmeal, soybean meal etc and they are promoted as an ideal ingredient in pigs, fish and poultry feed (**Figure 20**) or could be used as food for pet reptiles. However, it is aquaculture feed that offers the most promise as well as greatest potential return on investment due to the fact the fish feed is imported in most parts of Africa.

Initial research has been conducted using BSF meal as a partial or full substitute for fish or soybean meal in aquaculture diets. While many papers have reported good performance, others have found some slight decrease in growth rate, especially when the pre-pupae are used, which is attributed to the high level of chitin and significant reduction in digestibility. There is a clear need for more research on optimizing quality of the larvae as well as the processing method used. One advantage of BSF is that their dry matter content is relatively high—about 44%—compared to fishmeal, which may be about 25%. This should make drying and processing easier and cheaper.

It has been noted that supplementing larval food with fish renderings or algae can increase their omega-3 content, making them more suitable replacements for fishmeal. Total fat content is also strongly affected by diet. Reported values range from 15–21% of dry matter for larvae fed on poultry or cow manure, to 30% on a 50:50 mix of cow manure and fish offal to 42–49% when larvae are fed on oil rich food waste. This is far higher than fishmeal, which only contains 8–11% fat.

Another potential issue with BSF is that the outer skin contains chitin. Some fish, such as tilapia and turbot, are unable to easily digest chitin, so feeds containing this can reduce growth. Chitin is also a component of shellfish, so it could be expected that fish that naturally eat shrimp or crabs (or insects) would be able to process this compound. However, even among these species, lack of exposure during early development may mean the enzymes required are no longer present and growth is reduced.

These issues may be overcome through processing. Several options exist for handling the harvested pre-pupae. Initial studies simply used the harvested pre-pupae as they were – alive or dead, whole or chopped. However, this is only suitable for larger fish or reptiles. More commonly the BSF larvae are frozen, dried and ground into meal. Recent studies have mixed a meal with other ingredients in a pelletized form. If fat content needs to be reduced the frozen pupae can be cut before processing, allowing some of the intercellular fat to leak out. This fat can then be used for other purposes, such as biodiesel production.

A third option is to render the pupae (heating under pressure) to separate the protein, fat, chitin and other materials. Rendering allows these elements to be recombined in proportions optimized for a particular. For example, fishmeal is usually 60% protein or more, whereas BSF dry matter generally ranges from 40–44% protein. For this reason, a diet of BSF alone is inadequate for most fish species. BSF also contain more fat than fishmeal, and relatively high levels of dietary fibre.

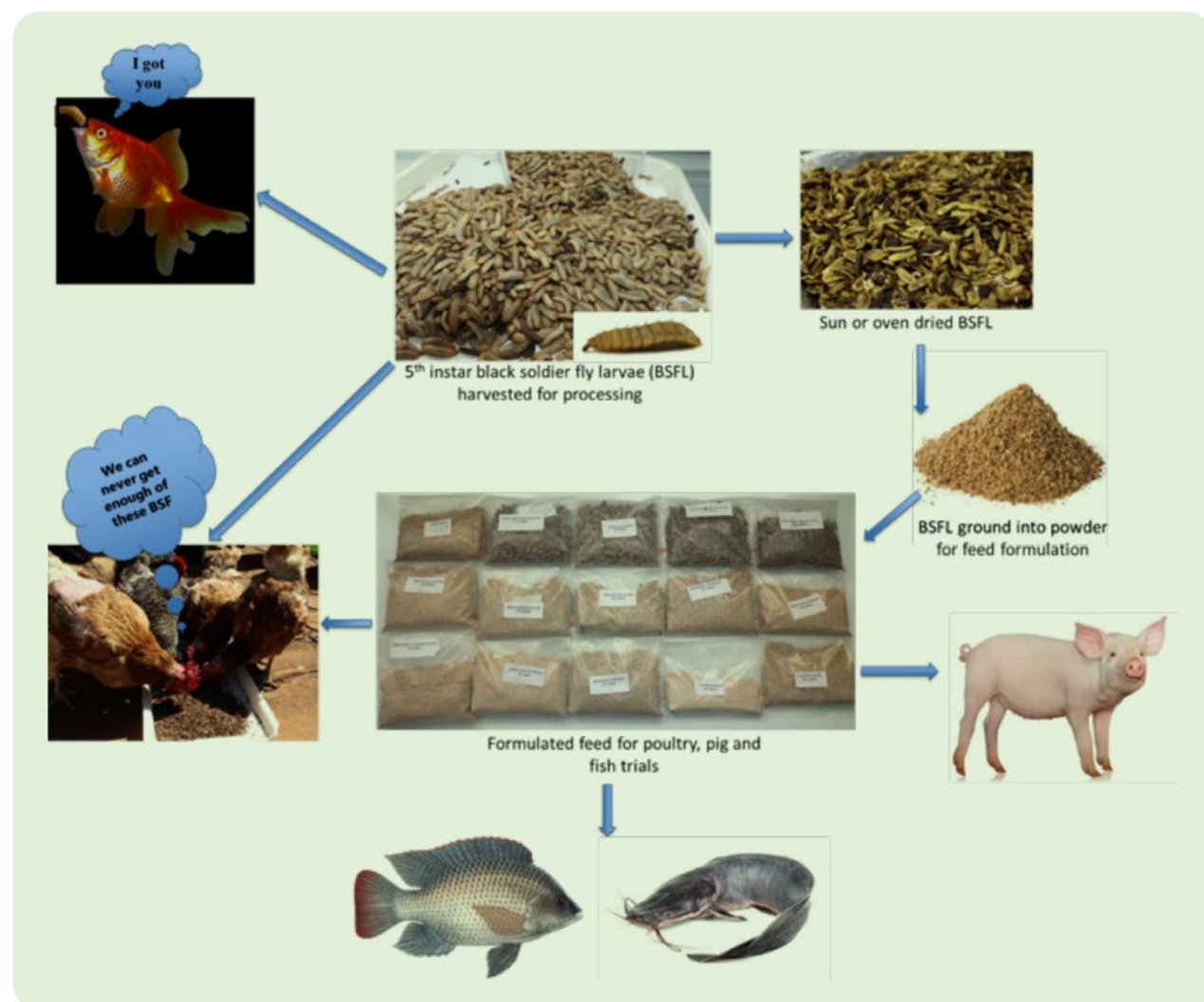


Figure 20: Formulated feeds for poultry, pig and fish

Organic fertilizer

The larvae are very efficient bioconvertors, and the residue which results from the digestion can be used as plant organic fertilizer and soil conditioner (**Figure 21**). The N-P-K value (Nitrogen –Phosphorous – Potassium) level is 5% – 3% – 2%, and as such is ideal for horticulture. The frass quality and quantity depends on the nature of the substrate and the period left inside the larvarium. The quality of the residue is sometimes considered to be too high in nutrients and it is often diluted. The organic matter is > 80%.



Figure 21: Residue of black soldier fly after bioconversion that could be transformed into organic fertilizer for enhanced soil fertility, high yield and growth of various plant types

Parasitoid, predators and pathogens of black soldier flies

The occurrence of the Entomopathogenic microbes *Apergillus flavus* (**Figure 22**), *Beauveria bassiana* (**Figure 23**) and *Metarhizium anisopliae* (**Figure 24**) and predatory mites (**Figure 25**) have been detected in Black soldier fly colonies, constituting a possible threat to mass rearing and colony growth but with efficient pre-and-postharvesting handling, these problems can be eliminated completely. Parasitic wasp *Dirhinus* sp. (Hymenoptera: Chalcididae) (**Figure 26**) and *Trichopria* sp. have been observed in the colonies attacking BSF pupae. These must be avoided at all costs. Adult insects should thus be kept in netted environments such as boxes and cages with access control, or structures, not only to prevent the from escaping, but also from preventing these predators to get to them.

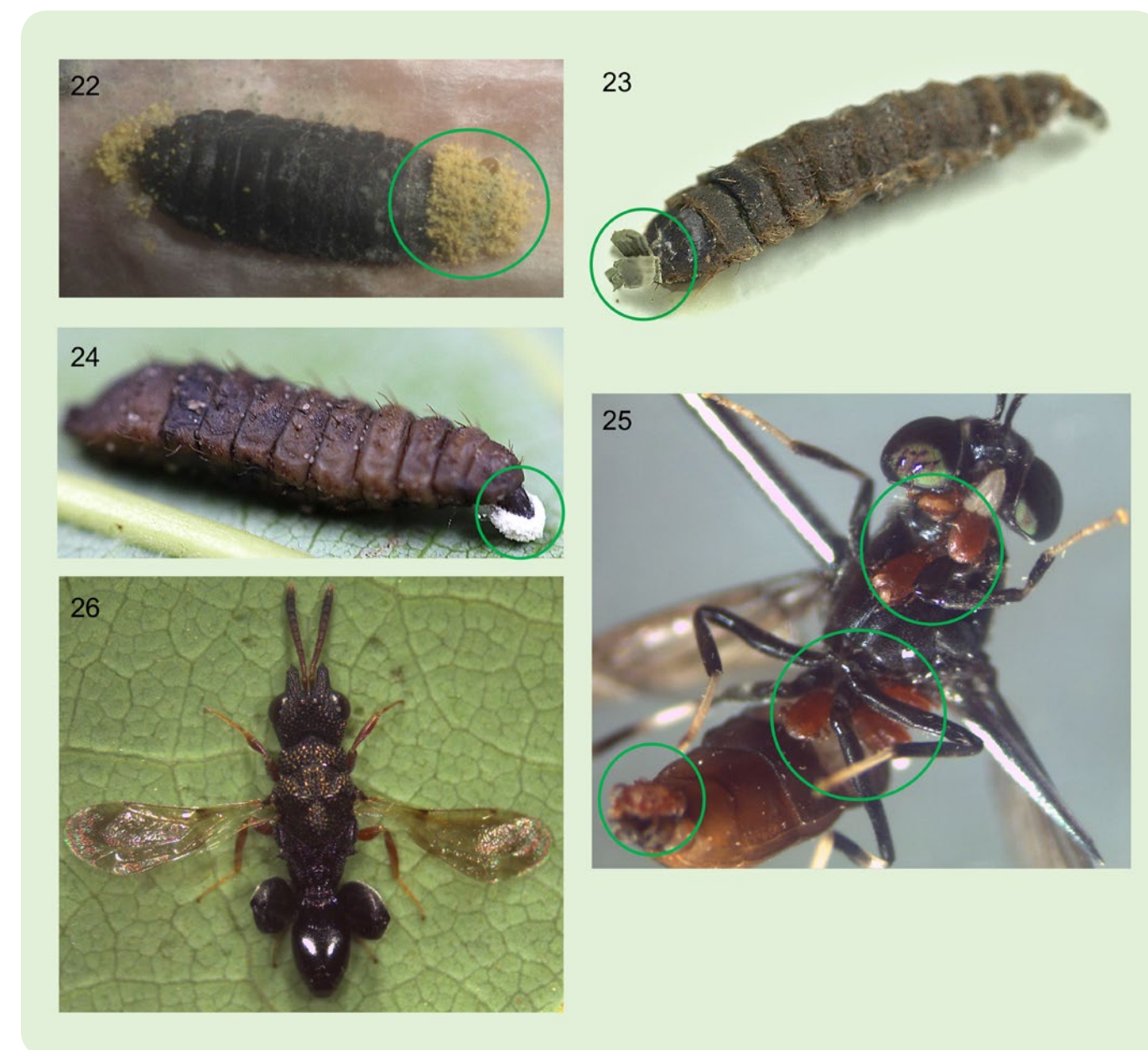


Figure 22, 23, 24, 25 and 26: Parasitoid, predators and pathogens of black soldier flies

Frequently Asked Questions (FAQS) about BSF

What is a black soldier fly?

A common fly (Diptera) of the family Stratiomyidae. Scientific name *Hermetia illucens*. It has the ability to transform various types of organic wastes into valuable proteins and organic fertilizer.

What does it look like?

BSF is black in colour and has similar size as a wasp except it has two wings, it doesn't bite nor does it constitute a nuisance.

Why should we learn about the Black Soldier's fly?

Pig, poultry and fish farming is constrained from realizing its full potential due to prohibitive cost of feeds, which constitute 60 – 70% of total cost of production. Commonly used protein sources include fish and plant protein sources, which are directly and indirectly competing with human nutrition. Therefore, farming BSF as an innovative cost-effective and eco-friendly options for alternative novel protein sources for livestock becomes crucial. BSF are extremely easy to rear and are rich in crude protein (35 – 50%), fatty acids, amino acids, vitamins, chitin, flavonoids etc which are very important in poultry, fish and pig nutrition.

How distributed is BSF?

It is present in many countries in the world including in Kenya.

What can they eat?

Adult BSF does not have a functional mouthpart as such they don't eat but they need water or sugar solution to extend their lifespan. However, the larvae are insatiable voracious feeders of organic waste – No limits, eats everything that begins to rot, especially kitchen waste, bran, market waste, spent grains from the breweries (brewery waste) and manure (pig, rabbit, goat, cow, chicken e.t.c).

Are they nuisance to humans?

The adults are not attracted to human habitats or foods and do not constitute a nuisance. They don't bite or transmit diseases. They spend most of their adult life mating and egg laying.

How long does a BSF live?

Minimum average lifespan of BSF is 6 days but sometimes they can live up to 16 – 40 days if maintained in control environment with water or supplied with 30 - 50% sugar solutions. Eggs take about 4 days to hatch, and then the larvae will take roughly 21 days to pupate. Pupae take 14 days to emerge as adult BSF's. Climatic factors do influence the lifespan of BSF.

How do we farm BSF?

BSF is farmed by simply rearing them on organic wastes: kitchen waste, bran, market waste, spent grains from the breweries (brewery waste) and manure (pig, rabbit, goat, cow, chicken etc) depending on the waste availability in each community. It is important to have a good insectarium or adult cage to maintain them for egg collection, good waste supply and good larvarium (for the larval stage rearing). In the larvarium, the larval stages should be maintained at different ages (staggered colony) to maintain a permanent stock of adults and eggs. The BSF colony should be maintained at $27.8 \pm 1^\circ\text{C}$, 60-70% Relative humidity and a photoperiod of 12 hours light (L12): 12 hours darkness (D12).

So, what's so good about BSF?

They convert various types of wastes to high-quality nutrients (especially protein), which largely depend on the rearing substrates and can be used as ingredient in feeds for poultry, fish and pig that are the fastest growing agribusinesses in Kenya providing income and employment opportunities for the youths and women.

What are the opportunities?

Mass production of BSF as a novel source of protein for poultry, fish and pig production in Kenya would lead to the creation of several new jobs for the youths and women in our communities.

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Project Title: Improving livelihood by increasing livestock production in Africa: An agribusiness model to commercially produce high quality insect-based protein ingredients for chicken, fish and pig industries (ILIPA)

Background information/Justification

The ILIPA project was developed by ICIPE in collaboration with partners from the Laboratory of Entomology, Wageningen University, The Netherlands to investigate the potential of insects in commercial production of low-cost, high-quality novel protein source to supplement feeds for poultry, pig and fish industries and the valorization of organic by-products by using insect species such as the black soldier fly (BSF), *Hermetia illucens* in Kenya.

Project goal is to establish profitable community-led agro-enterprises for commercial production of safe and high-quality insect-based feeds for the poultry, pig and fish farmers to improve livelihood and promote youths and women employment through capacity building on the use of insects as a novel source of proteins in feeds for livestock production (pig, poultry and fish). The project will work with youth and women farmer groups to build their capacity to rear BSF and market thereby ensuring farmer participation in establishment of intensive insect-based agribusiness enterprises. The project will also engage in research to assure high nutrition and microbial safety of the insect-based protein products.



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